

Findings and Evidence: Platform Membership as Marketing-Mix Moderator

This table summarizes the interpretation-relevant findings from the clean analysis module. The main outcome is transformed capacity utilisation (`ur_boxcox`). The main fixed-effect strategies are `market_fe` (city, date, and operator fixed effects) and `unit_fe` (city-operator and date fixed effects). Standard errors are clustered by `city_operator`. The referenced files are all included in this repository.

Table 1: Relevant findings, empirical reference values, and interpretation.

Finding	Reference value	Interpretation sentence
No robust universal platform bonus	platform_any : market-FE coef = 0.0385, $p = 0.0468$; unit-FE coef = 0.0040, $p = 0.788$	Platform membership should not be framed as an unconditional utilisation booster because the positive market-FE association disappears once city-operator fixed effects are used.
Platform type is empirically heterogeneous	Mean ur : none = 0.00755, FreeNow-only = 0.00845, local MaaS-only = 0.01775, multi-platform = 0.00816, weak exposure-only = 0.00691	The data support treating platform membership as a set of distinct platform contexts rather than a homogeneous binary state.
Local MaaS has the highest descriptive utilisation	Local MaaS-only: 1,686 rows, 18 city-operators, mean ur = 0.01775 versus none = 0.00755	Local MaaS integrations appear descriptively associated with substantially higher capacity utilisation, but causal interpretation still requires the moderation models.
Weak exposure is not equivalent to membership	Weak exposure-only: 305 rows, 2 city-operators, mean ur = 0.00691; unit-FE main term coef = -0.0617 , $p < 0.001$	Weak exposure should be separated from real bookable/payable platform membership because it behaves empirically like a different condition.
Core moderation: access pricing friction	unlock_fee $\times$ any_platform : market-FE coef = -0.0715 , $p < 0.001$; 9 negative significant estimates across 16 model variants	The strongest story is that platform membership changes marketing effectiveness by making access-pricing frictions more harmful for capacity utilisation.
Local MaaS amplifies unlock-fee friction	unlock_fee $\times$ local_maas_only : market-FE coef = -0.1075 , $p < 0.001$; 9 negative significant estimates across 16 variants	The pricing-friction result is especially pronounced in local MaaS contexts, suggesting that integrated mobility platforms may heighten sensitivity to fixed access costs.
Unlock fee is more central than per-minute price	unlock_fee $\times$ any_platform : coef = -0.0715 , $p < 0.001$; price/min $\times$ any_platform : coef = -0.0049 , $p = 0.818$ in the same market-FE model	The relevant pricing mechanism is not the variable usage price but the fixed access friction that users face before a trip starts.
FreeNow-only is not a robust positive performance context	large_aggregator_only : market-FE main coef = -0.0461 , $p = 0.0559$; unit-FE main coef = 0.0006, $p = 0.970$	FreeNow-only membership does not provide a stable direct capacity-utilisation advantage once richer fixed effects are applied.
FreeNow promotion effect is suggestive but not robust	promotion $\times$ FreeNow-only : market-FE coef = 0.1304, $p < 0.001$; unit-FE coef = 0.0002, $p = 0.987$	There is suggestive evidence that promotions work better in the FreeNow context, but it should be interpreted cautiously because it is not robust to city-operator fixed effects.
Coverage effects are not universally platform-enhanced	coverage $\times$ any_platform : market-FE coef = -0.0245 , $p = 0.112$; unit-FE coef = 0.0014, $p = 0.862$	Platform membership does not generally make simple service coverage more effective for capacity utilisation.
Multi-platform coverage has a unit-FE signal	coverage $\times$ multi_platform : unit-FE coef = 0.0516, $p = 0.006$; market-FE coef = -0.0045 , $p = 0.840$	Coverage may matter more in multi-platform contexts within the same city-operator unit, but this result is narrower than the unlock-fee finding.

Finding	Reference value	Interpretation sentence
Exclusive coverage can turn negative in platform contexts	<code>exclusive_coverage</code> $\times$ <code>FreeNow-only</code> : market-FE coef = -0.0483 , $p = 0.0065$; local MaaS unit-FE coef = -0.0731 , $p < 0.001$; multi-platform unit-FE coef = -0.0664 , $p < 0.001$	Exclusive territorial coverage is not consistently beneficial under platform membership and may reflect lower-comparison or lower-demand contexts.
Competition mechanism is plausible but exploratory	<code>unlock_fee</code> $\times$ <code>any_platform</code> $\times$ <code>competition</code> : market-FE coef = -0.0330 , $p = 0.099$; no-weak-exposure market-FE coef = -0.0394 , $p = 0.060$	Competition provides a plausible mechanism for the pricing-friction story, but the evidence is not stable enough to serve as the main claim.
DML partially supports the mechanism but is not decisive	HGB-DML no-weak-exposure: <code>unlock_fee</code> $\times$ <code>local_only</code> $\times$ <code>competition</code> coef = -0.0353 , $p = 0.0057$; RF-DML signs are not fully consistent	Double Machine Learning offers some support for competition-amplified pricing friction, but estimator sensitivity means the DML evidence should be framed as robustness probing rather than the core result.
Capacity utilisation is the appropriate performance lens	Primary outcome is <code>ur_boxcox</code> ; robustness outcome is <code>log_trip_count_day</code> ; all main tables report 14,514 all-sample observations and 97 city-operators	The contribution is about provider-side efficiency of deployed capacity, not only about demand volume or trip counts.

Naive interpretation

A naive reading of platform membership would suggest that joining a platform should improve provider performance by expanding reach, increasing visibility, and giving providers access to a larger pool of potential users. Our results do not support this simple demand-expansion interpretation. Instead, the evidence suggests that platform membership changes the conditions under which marketing instruments translate into capacity utilisation. In particular, platform contexts appear to make fixed access-pricing frictions more consequential: unlock fees are more negatively associated with capacity utilisation when providers are embedded in platform-mediated markets, and this pattern is especially visible for local MaaS integrations. This implies that platforms do not merely add demand; they also reshape the choice environment in which users compare providers and evaluate the cost of starting a trip.

The paper-worthy interpretation is therefore that platform membership alters the effectiveness of the marketing mix rather than producing a universal performance premium. Providers cannot treat platforms as neutral reach channels. Once a provider becomes visible or bookable through a platform, access costs such as unlock fees become more salient and may reduce the efficiency with which deployed capacity is used. Local MaaS platforms appear to be the clearest setting for this mechanism, while FreeNow-only membership does not show a robust direct utilisation advantage. The contribution is thus to shift the provider-side platform discussion from whether platform membership pays off on average to when and through which marketing instruments it changes provider performance.